EE 121B: Principles of Semiconductor Device Design

Lecture 8:

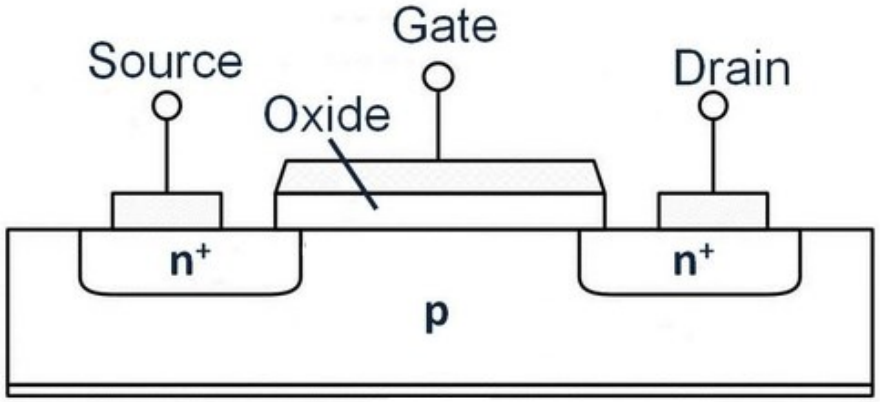
pn Junction Diodes

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. Metal-Semiconductor Contacts
3. P-N Junction
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



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Lecture 8

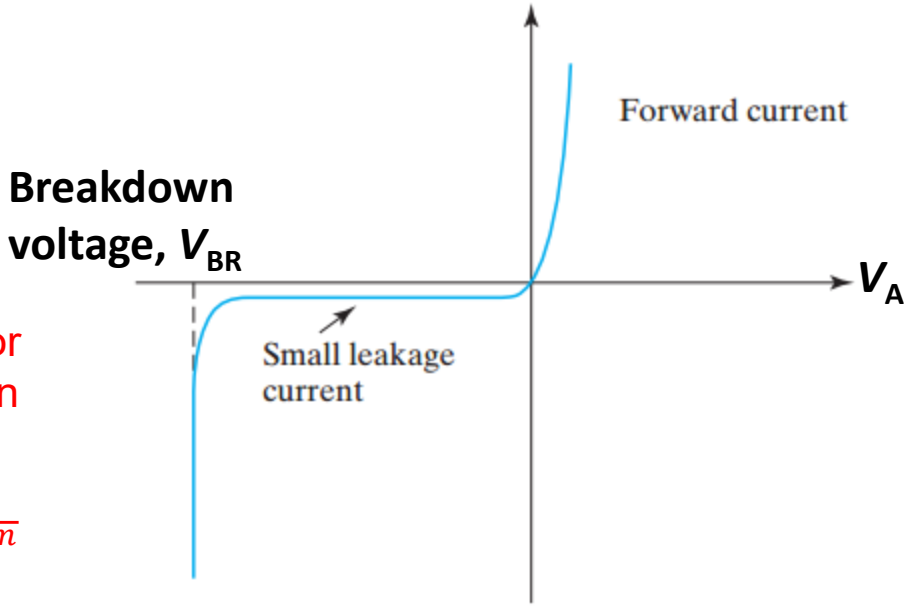
OUTLINE

- pn Junction Diodes (cont'd)
  - Junction breakdown
  - Deviations from the ideal I-V
    - R-G current
    - series resistance
    - high-level injection

Reading: Pierret 6.2; Hu 4.5

pn Junction Breakdown

C. C. Hu, *Modern Semiconductor Devices for Integrated Circuits*, Figure 4-10



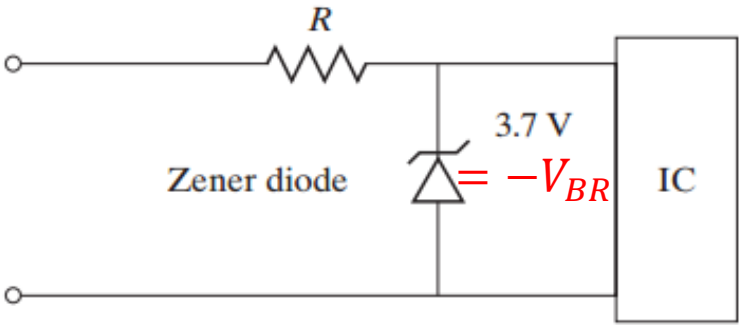
Conducts a lot of current if the voltage reaches up to 3.7V → protects IC

Empirical model for reverse breakdown current:

$$I = \frac{I_0}{1 - \left|\frac{V_A}{V_{BR}}\right|^m}$$

$$3 < m < 6$$

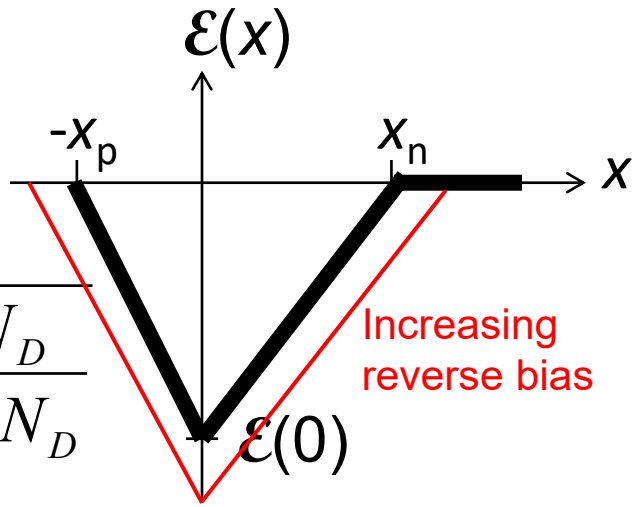
A ***Zener diode*** is designed to operate in the breakdown mode:



Review: Peak $\mathcal{E}$-Field in a pn Junction

$$\mathcal{E} = \int \frac{\rho}{\varepsilon_{Si}} dx$$

$$\left|\mathcal{E}(0)\right| = \frac{qN_A x_p}{\varepsilon_{Si}} = \frac{qN_D x_n}{\varepsilon_{Si}} = \sqrt{\frac{2q\left(V_{bi} - V_A\right)}{\varepsilon_{Si}} \frac{N_A N_D}{N_A + N_D}}$$



For a one-sided junction,

$$\left|\mathcal{E}(0)\right| \cong \sqrt{\frac{2q\left(V_{bi} - V_A\right)N}{\varepsilon_{Si}}}$$

where *N* is the dopant concentration on the lightly doped side

Breakdown Voltage, V_{BR}

- If the reverse bias voltage ($-V_A$) is so large that the peak electric field exceeds a critical value $\mathcal{E}_{CR}$, then the junction will "break down" (*i.e.* large reverse current will flow)

$$\mathcal{E}_{CR} = \sqrt{\frac{2qN\left(V_{bi} + V_{BR}\right)}{\varepsilon_s}} \quad = \mathcal{E}(0) \text{ when } V_A = -V_{BR}$$

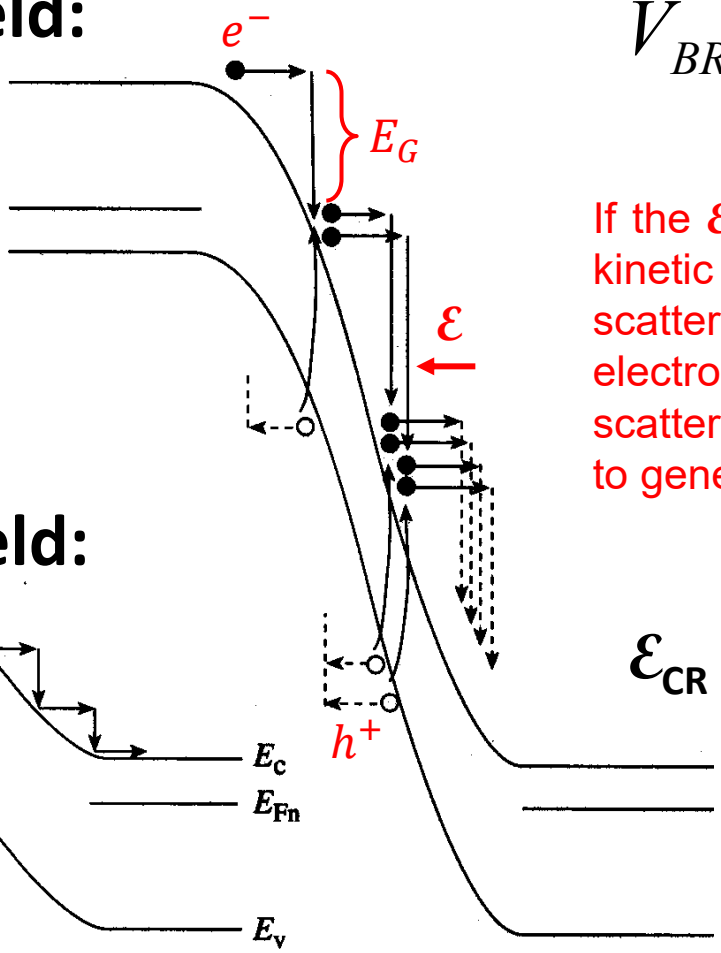
- Thus, the reverse bias at which breakdown occurs is

$$V_{BR} = \frac{\varepsilon_s \mathcal{E}_{CR}^{\ 2}}{2qN} - V_{bi}$$

Avalanche Breakdown Mechanism

R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 6.12

High $\mathcal{E}$-field:



Low $\mathcal{E}$-field:

$$V_{BR} \approx \frac{\varepsilon_s \mathcal{E}_{CR}^{\ 2}}{2qN} \quad \text{if } V_{BR} >> V_{bi}$$

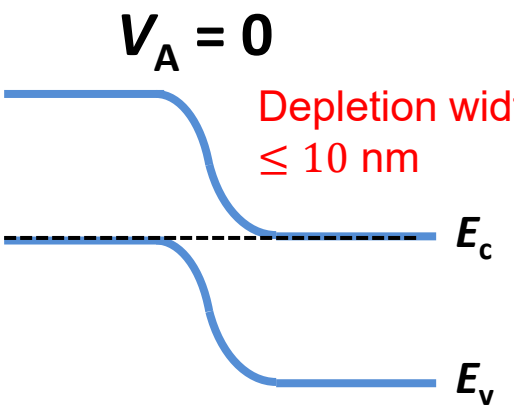
If the $\mathcal{E}$ field is sufficiently large such that the kinetic energy that an electron gains between scattering events exceeds E_G, then an electron-hole pair can be generated upon scattering. Each generated carrier can go on to generate additional electron-hole pairs.

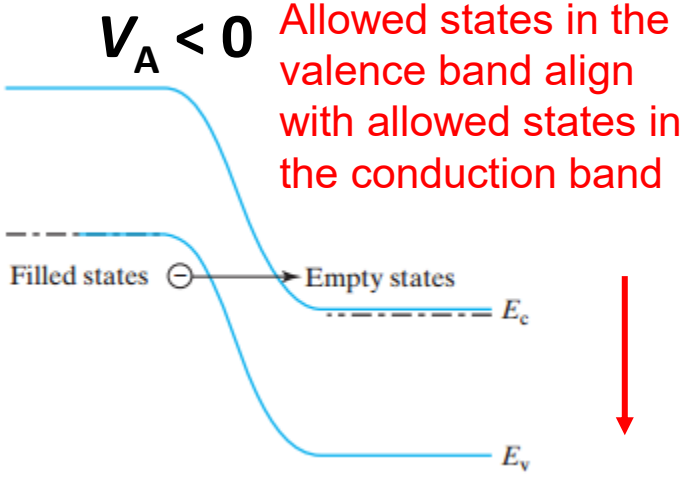
$\mathcal{E}_{CR}$ increases slightly with *N*:

For 10^{14} cm^{-3} < *N* < 10^{18} cm^{-3},
10^5 V/cm < $\mathcal{E}_{CR}$ < 10^6 V/cm

Tunneling (Zener) Breakdown Mechanism

Dominant breakdown mechanism when both sides of a junction are very heavily doped.





$$V_{BR} = \frac{\varepsilon_s \mathcal{E}_{CR}^{\ 2}}{2qN} - V_{bi}$$

$$\mathcal{E}_{CR} \approx 10^6 \text{ V/cm}$$

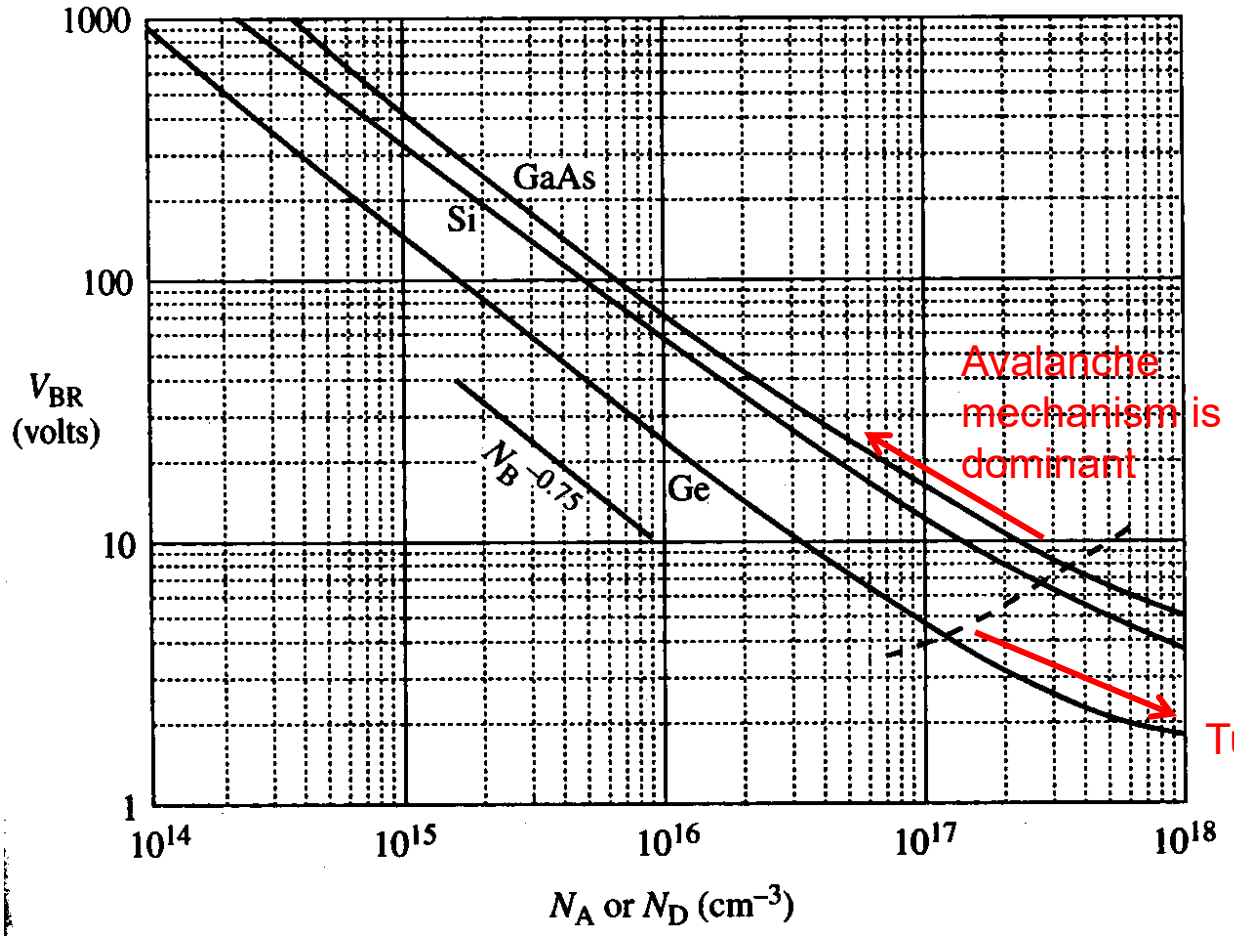
Significant quantum mechanical tunneling occurs if width is very narrow (<10nm)

Typically, $V_{BR} < 5$ V for Zener breakdown

C. C. Hu, *Modern Semiconductor Devices for Integrated Circuits*, Figure 4-12

Empirical Observations of V_{BR}

R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 6.11

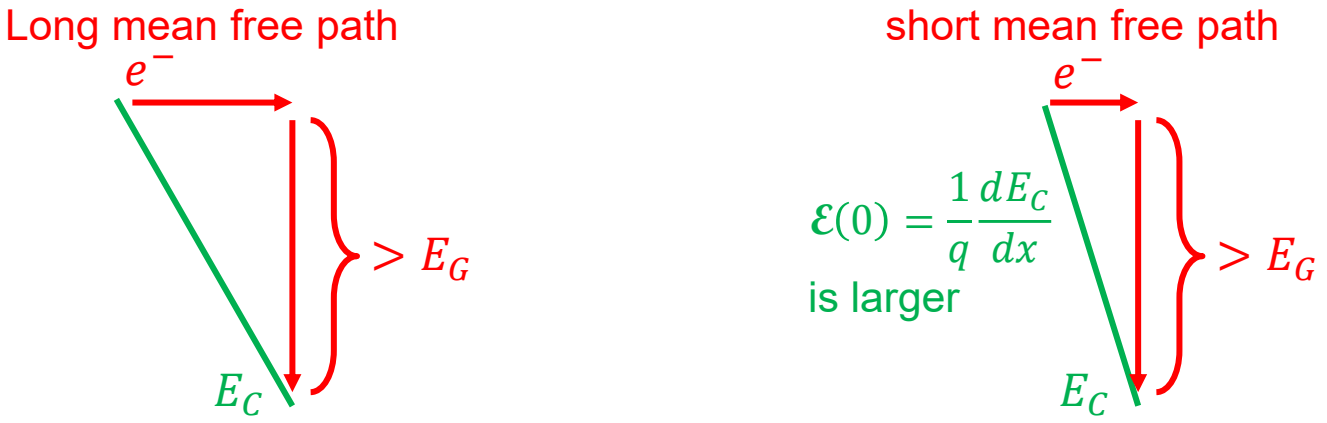


$$E_{G,GaAs} = 1.42\text{eV}$$
$$E_{G,Si} = 1.12\text{eV}$$
$$E_{G,Ge} = 0.67\text{eV}$$

- V_{BR} decreases with increasing N
- V_{BR} decreases with decreasing E_G

V_{BR} Temperature Dependence

- For the **avalanche mechanism**:
  - V_{BR} *increases* with increasing *T*, because the mean free path decreases



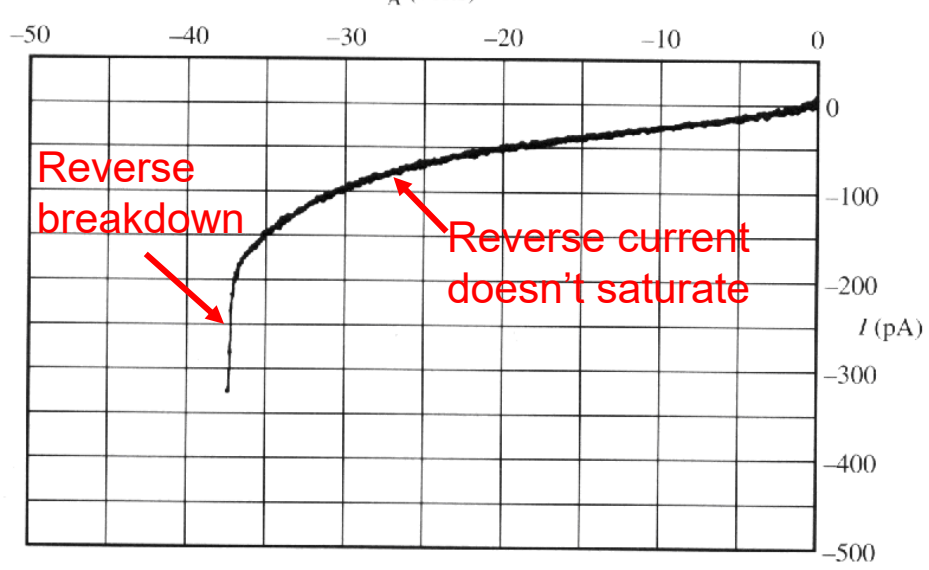
- For the **tunneling mechanism**:
  - V_{BR} *decreases* with increasing *T,* because the flux of valence-band electrons available for tunneling increases

Deviations from the Ideal *I-V*

R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 6.10

Reverse-Bias Current

(linear scale)



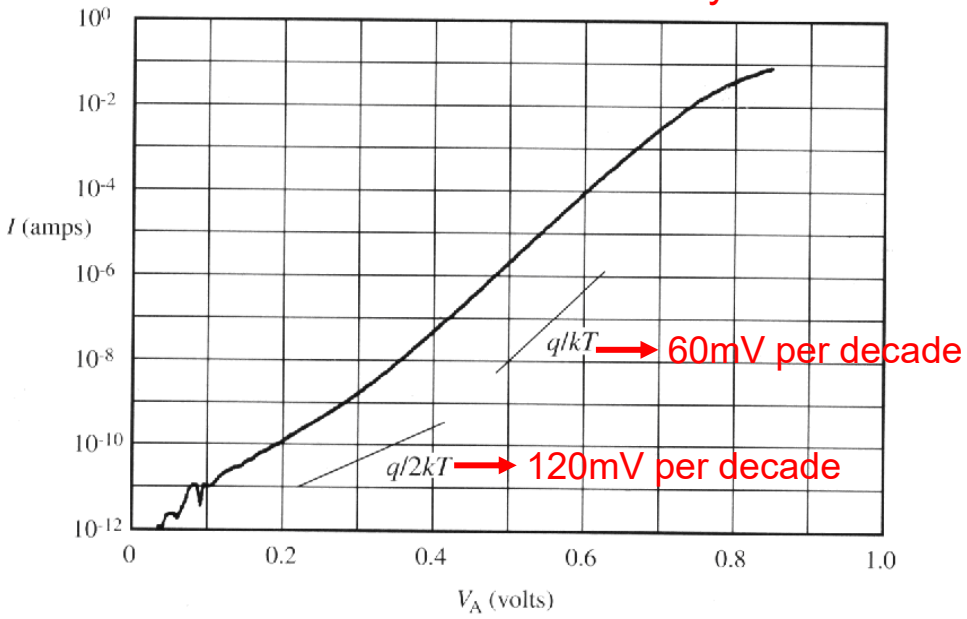
Ideally, $I \cong -I_0 = \text{constant}$

Forward-Bias Current

(log scale)

$I \cong I_0 e^{qV_A/nkT}$

n is ideality factor ≥ 1



Ideally,

$$\log(I) \cong \log(I_0) + \log\left(e^{qV_A/kT}\right)$$

$$= const. + \left(qV_A/kT\right)\log(e) = const. + \frac{V_A}{(kT/q)\ln(10)}$$

Effect of R-G in Depletion Region

- The net generation rate is given by

$$\frac{\partial p}{\partial t} = \frac{\partial n}{\partial t} = \frac{n_i^{\,2} - np}{\tau_p (n + n_1) + \tau_n (p + p_1)}$$

where $n_1 \equiv n_i e^{(E_T - E_i)/kT}$ and $p_1 \equiv n_i e^{(E_i - E_T)/kT}$

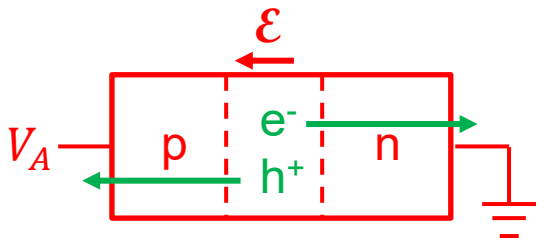
E_T = trap-state energy level

There is net generation of electron-hole pairs (EHP) if $pn < n_i^2$
e.g. in the depletion region and under reverse bias
There is net recombination of EHP if $pn > n_i^2$
e.g. in the depletion region under forward bias

- R-G in the depletion region contributes an additional component of diode current $I_{R\text{-}G}$:

$$I_{R-G} = -qA \int_{-x_p}^{x_n} \left. \frac{\partial p}{\partial t} \right|_{R-G} dx$$

Generation of the EHP results in one mobile charge carrier passing through the diode



Net Generation in Reverse Bias

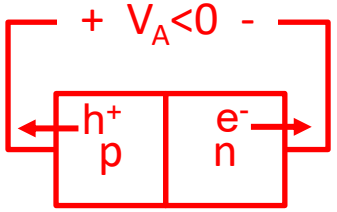
R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 6.15(a)

- For **reverse bias** greater than several kT/q,

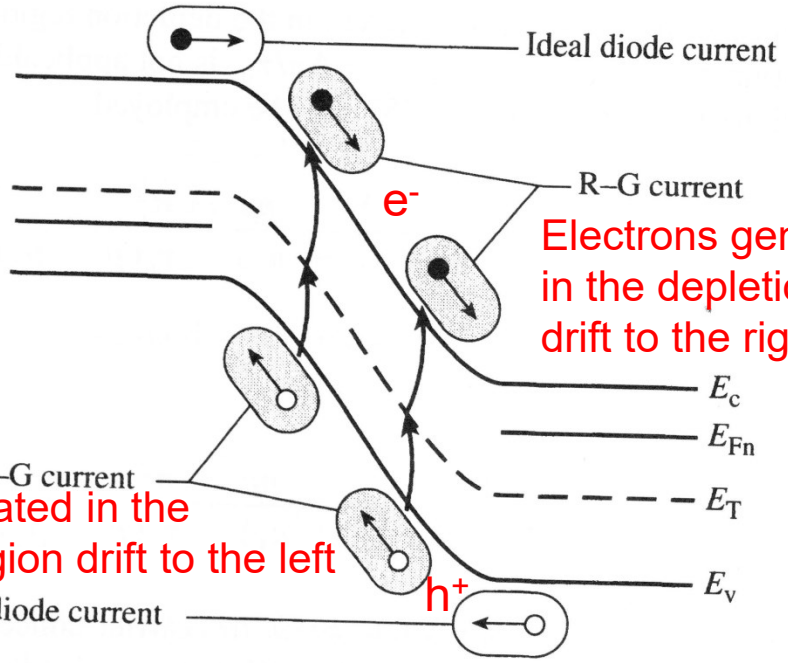
Increases with reverse bias voltage

$$I_{R-G} = -\frac{qAn_iW}{2\tau_0} \quad \text{where} \quad \tau_0 \equiv \frac{1}{2}\left(\tau_p \frac{n_1}{n_i} + \tau_n \frac{p_1}{n_i}\right)$$

Negative current



Majority carriers flow out of the quasi-neutral regions



Collection of minority carriers which diffuse into the depletion region

Electrons generated in the depletion region drift to the right

Holes generated in the depletion region drift to the left

Resulting in an extra component of negative current

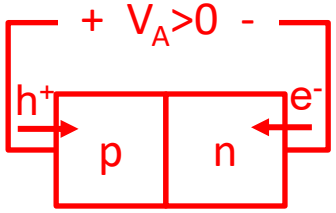
Net Recombination in Forward Bias

R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 6.15(b)

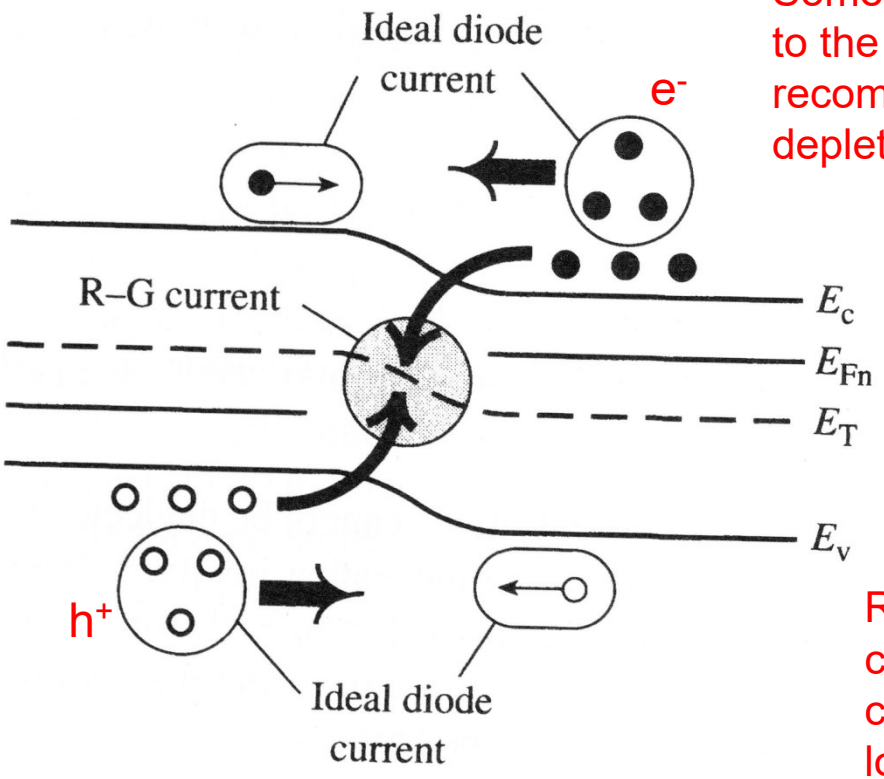
- For **forward bias:**

n=2

$$I_{R-G} \propto qAn_i We^{qV_A/2kT}$$



Majority carriers flow into of the quasi-neutral region



Some electrons supplied to the n-type region recombine in the depletion region

Some holes supplied to the p-type region recombine in the depletion regions

Resulting in an extra component of positive current (dominant at low V_A)

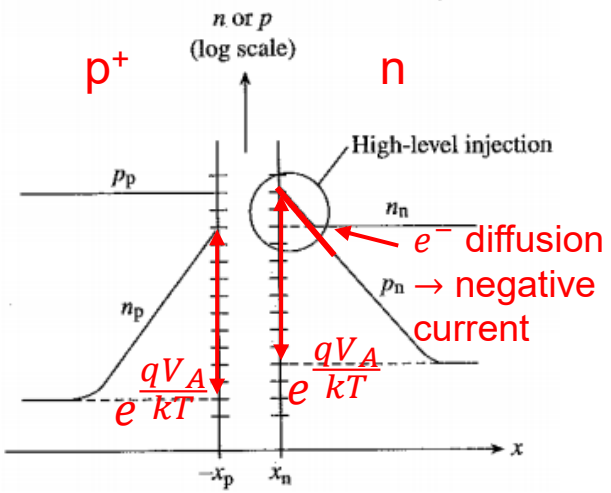
High-Level Injection (HLI) Effect

- As V_A increases, the side of the junction which is more lightly doped will eventually reach HLI:

$n_n > n_{no}$ for a p^+n junction

or

$p_p > p_{po}$ for a pn^+ junction



$\Rightarrow$ significant gradient in majority-carrier profile

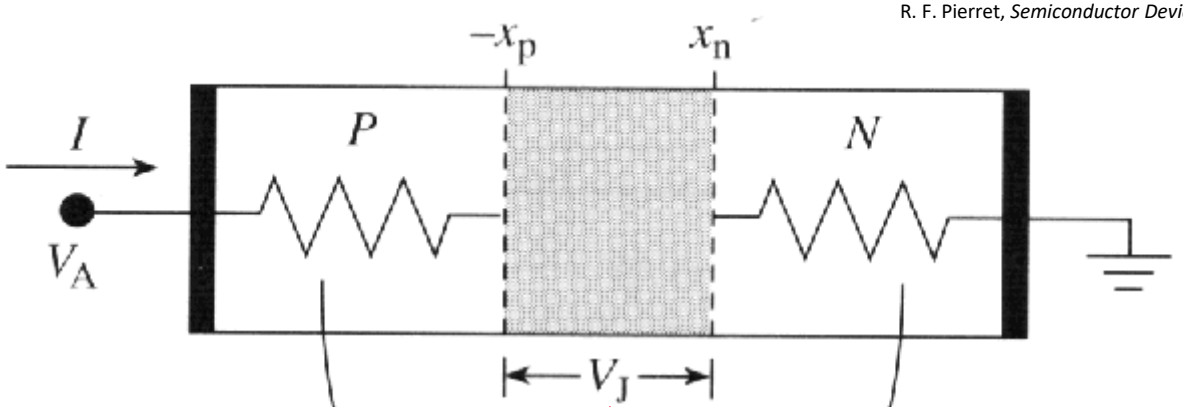
(Majority-carrier diffusion current is no longer negligible)

Majority-carrier diffusion current reduces the diode current from the ideal case.

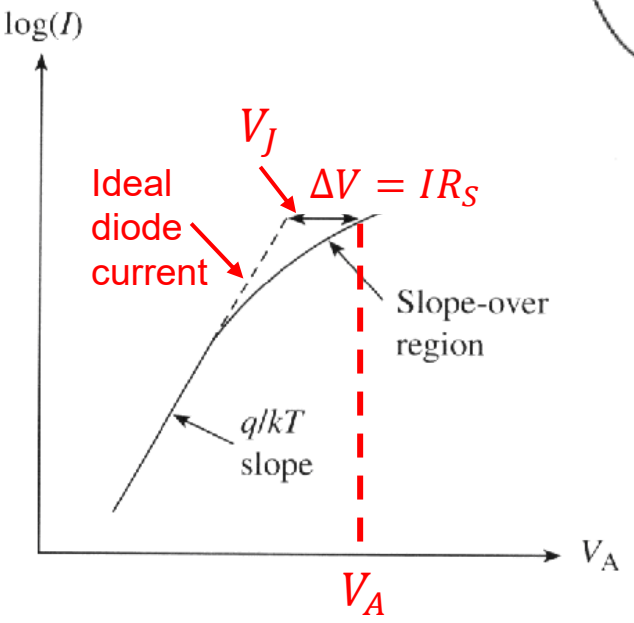
R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 6.17(a)

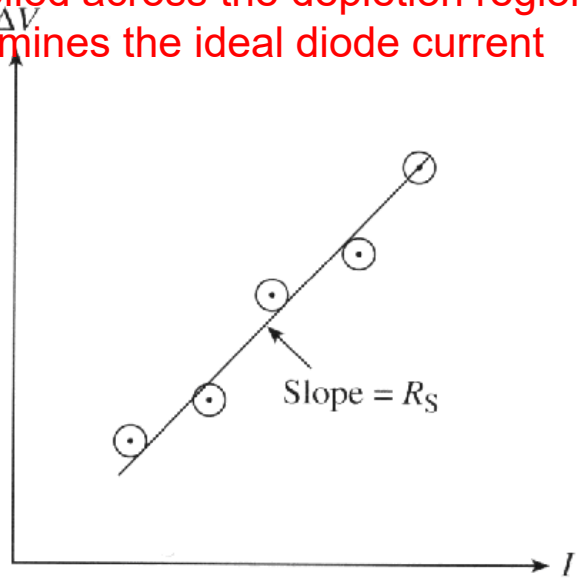
Effect of Series Resistance

R. F. Pierret, *Semiconductor Device Fundamentals*, Figure 6.16



Voltage applied across the depletion region, which determines the ideal diode current





Summary: Junction Breakdown

- If the peak electric field in the depletion region exceeds a critical value $\mathcal{E}_{CR}$, then large reverse current will flow.

  This occurs at a negative bias voltage called the breakdown voltage, V_{BR}:

$$V_{BR} = \frac{\varepsilon_s \mathcal{E}_{CR}^{\ 2}}{2qN} - V_{bi}$$

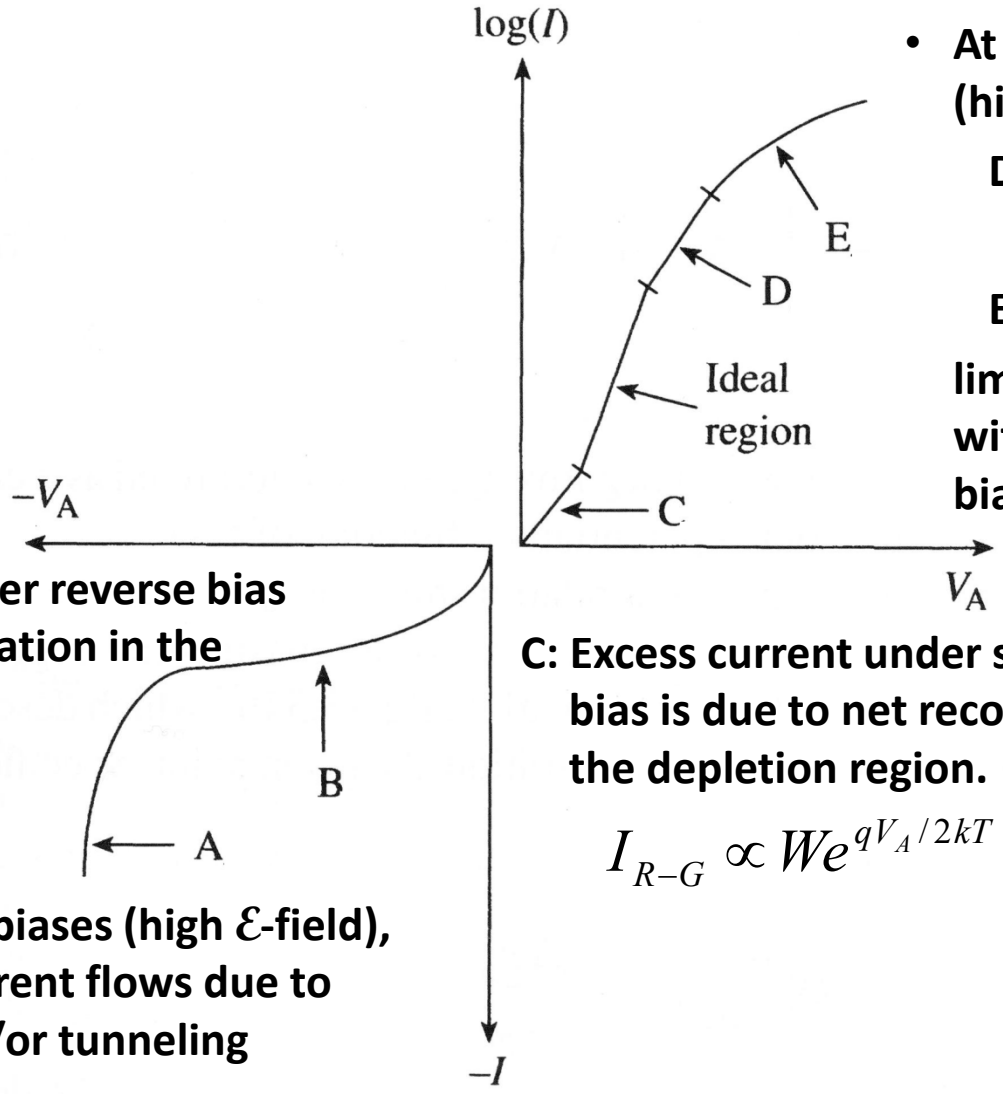
  where N is the dopant concentration on the more lightly doped side

- The dominant breakdown mechanism is

  avalanche, if $N < \sim 10^{18}/cm^3$

  tunneling, if $N > \sim 10^{18}/cm^3$

Summary: Deviations from Ideal *I-V*



- **At large forward biases (high current densities)**
  - **D: high-level injection**
    $\rightarrow I \propto e^{qV_A/2kT}$
  - **E: series resistance**

limit increases in current with increasing forward bias voltage.

B: Excess current under reverse bias is due to net generation in the depletion region.

$|I_{R-G}| \propto W$

C: Excess current under small forward bias is due to net recombination in the depletion region.

$I_{R-G} \propto We^{qV_A/2kT}$

A: At large reverse biases (high $\mathcal{E}$-field), large reverse current flows due to avalanching and/or tunneling

R. F. Pierret, *Semiconductor Device Fundamentals*, Figure E6-9

Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes